Creative Programming: Sound

Lecture 5 (24 Feb)

Narration

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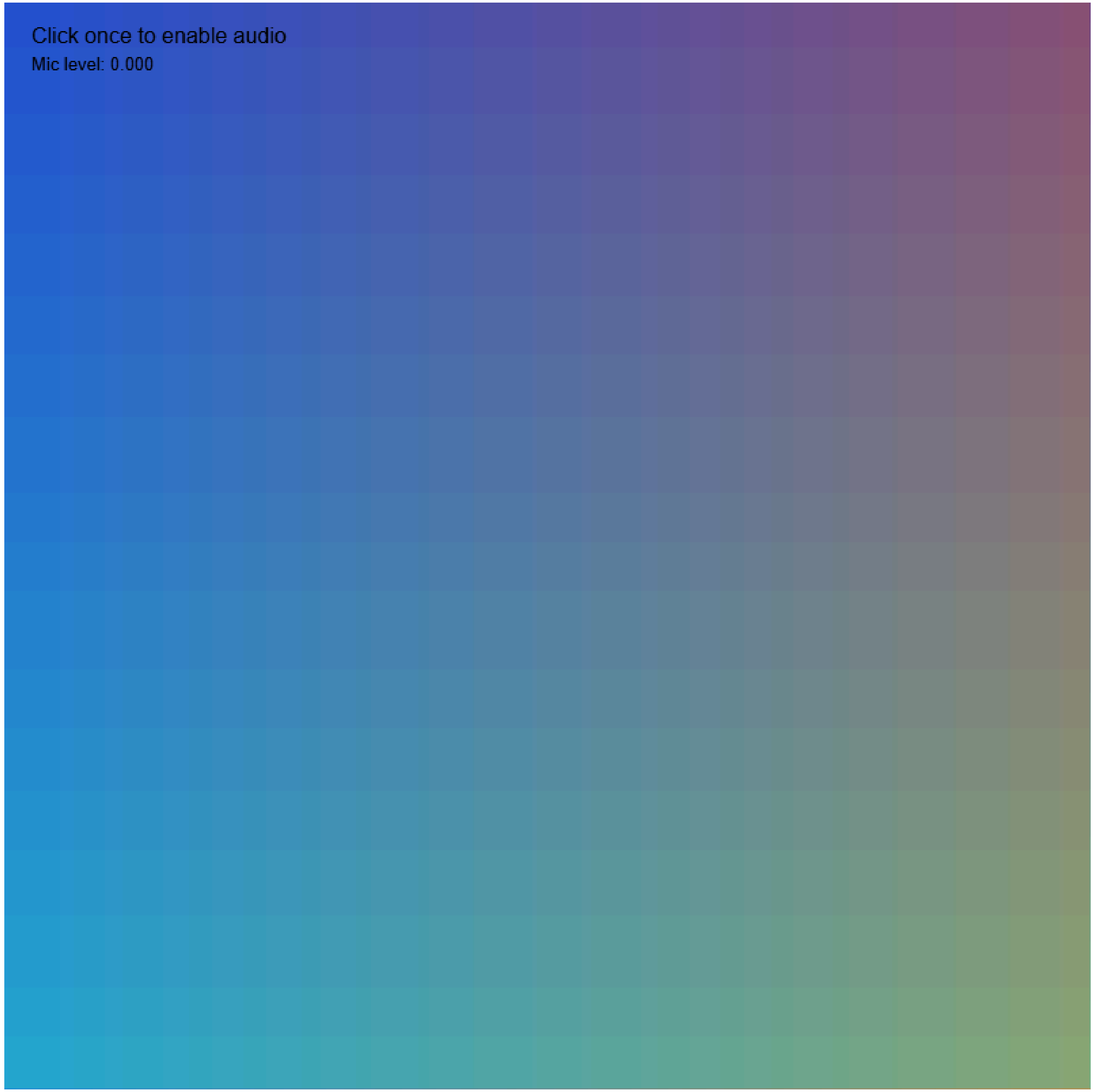

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(Re)visiting slides?

You can jump to any slide with the menu to the (bottom) left.

Did you miss a slide or want to revisit?
Open the narration tab while
studying to get an explanation of
difficult slides.

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Milestone 1:

- We will go through the requirements for the first milestone, which is due next week

Milestone Guidance

- If you need some feedback/help for the first milestone, contact me during one of the class assignments or breaks
- As always, you can also ask and discuss with the TAs on Thursdays

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Bounding Values

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Constraining via if

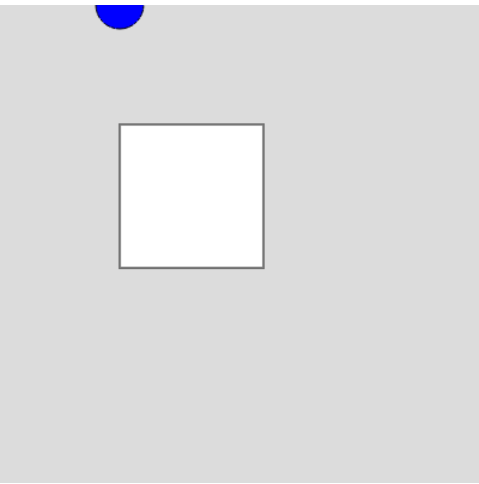
- if statements can be used to constrain values, but they can be verbose and error-prone.
- Example: constraining a variable `constrainedX` to the square.

```
function setup() {
  createCanvas(400, 400);
}

function draw() {
  background(220);
    fill("white");
  rect(100, 100, 120, 120); // Box in the background

  let constrainedX = mouseX; // We need to manipulate the value of mouseX, so we copy it to a new variable

  if (constrainedX < 100) { // Constrain left side
    constrainedX = 100;
```



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Constraining via constrain

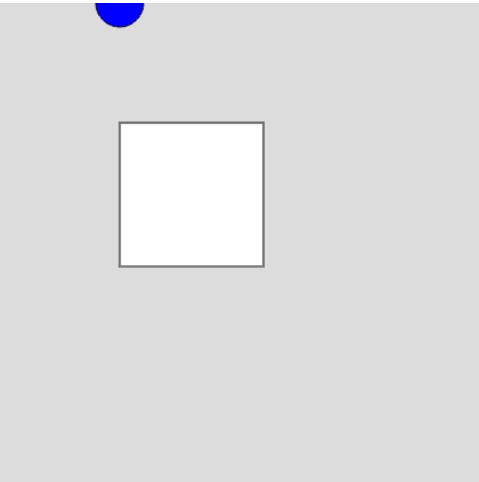
- p5.js provides a built-in function `constrain()` that simplifies the process of constraining values.
- Example: constraining a variable `constrainedX` using `constrain()`.

```
function setup() {
  createCanvas(400, 400);
}

function draw() {
  background(220);
  fill("white");
  rect(100, 100, 120, 120); // Box in the background

  let constrainedX = constrain(mouseX, 100, 220); // Constrain mouseX between 100 and 220

  fill("blue");
  circle(constrainedX, mouseY, 40); // Draw circle on constrained position
}
```



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Constraining via min/max

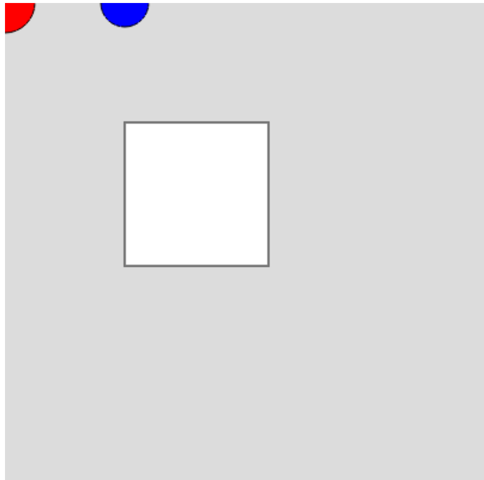
- Another way to constrain values is by using the `min()` and `max()` functions.
- `min()` returns the smaller of two values, while `max()` returns the larger.
- Example: constraining a variable `constrainedX` using `min()` and `max()`.

```
function setup() {
  createCanvas(400, 400);
}

function draw() {
  background(220);
  fill("white");
  rect(100, 100, 120, 120); // Box in the background

  let constrainedX = min(mouseX, 220); // Constrain right side (100 + 120)

  fill("red");
  circle(constrainedX, mouseY, 50); // Draw circle with right side constrained
```



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Constraining via modulo

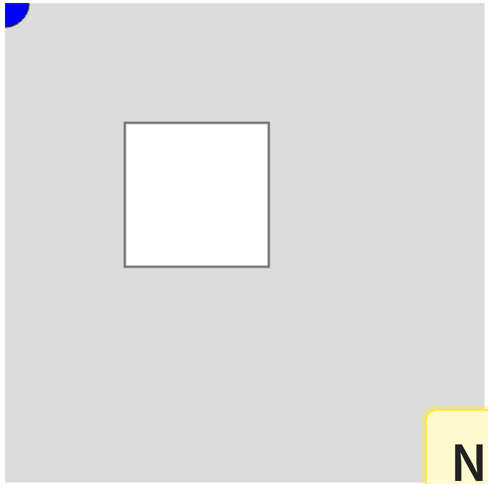
- The modulo operator `%` can be used to wrap values around a certain range.
- This is particularly useful for creating looping behavior, such as in animations or games.
- Example: wrapping a variable `moduloX` to be between 0 and 220.

```javascript
function setup() {
  createCanvas(400, 400);
}

function draw() {
  background(220);
  fill("white");
  rect(100, 100, 120, 120); // Box in the background

  let moduloX = mouseX % 220; // Modulo to wrap mouseX

```



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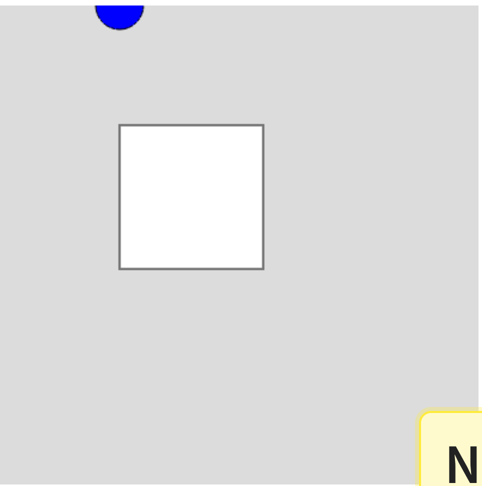
Constraining via mapping

- The `map()` function scales and constrains values.
- You map two ranges to each other, and the function will automatically scale the input value to fit within the output range.
- Example: mapping a variable `mappedX` from 0-400 to 100-220.

```
function setup() {
  createCanvas(400, 400);
}

function draw() {
  background(220);
  fill("white");
  rect(100, 100, 120, 120); // Box in the background

  let mappedX = map(mouseX, 0, width, 100, 220); // Map full canvas mouseX range to square x range
```



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Constraining in action

Class Assignment: Car

1. Create a shape that moves with the arrow keys. (Tip: check the reference!)
2. Use one of the constraining techniques to keep the shape within the canvas boundaries.
3. Can you make it so that the shape wraps around the canvas instead of stopping at the edges?

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The World of Sound

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Sound in the real world

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Paint with Music

Paint with Music by Simon Doury and Caroline Buttet

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Coded Instruments

Narration

The synthesizer

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The p5.js synthesizer

- We can play sound in p5.js using a synthesizer

```
let synthesizer;

function setup() {
  createCanvas(400, 400);
  synthesizer = new p5.MonoSynth()
}

function draw() {
  background(220);
  text("Click me to play 'tuning A'", 100, height / 2);

}

function mousePressed() {
  userStartAudio();
  synthesizer.play("A4", 1, 0, 1);
}
```

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Libraries

- This code does not work by default! We need to include a library for extra functionality.
- The p5.js sound library provides various tools for working with sound.

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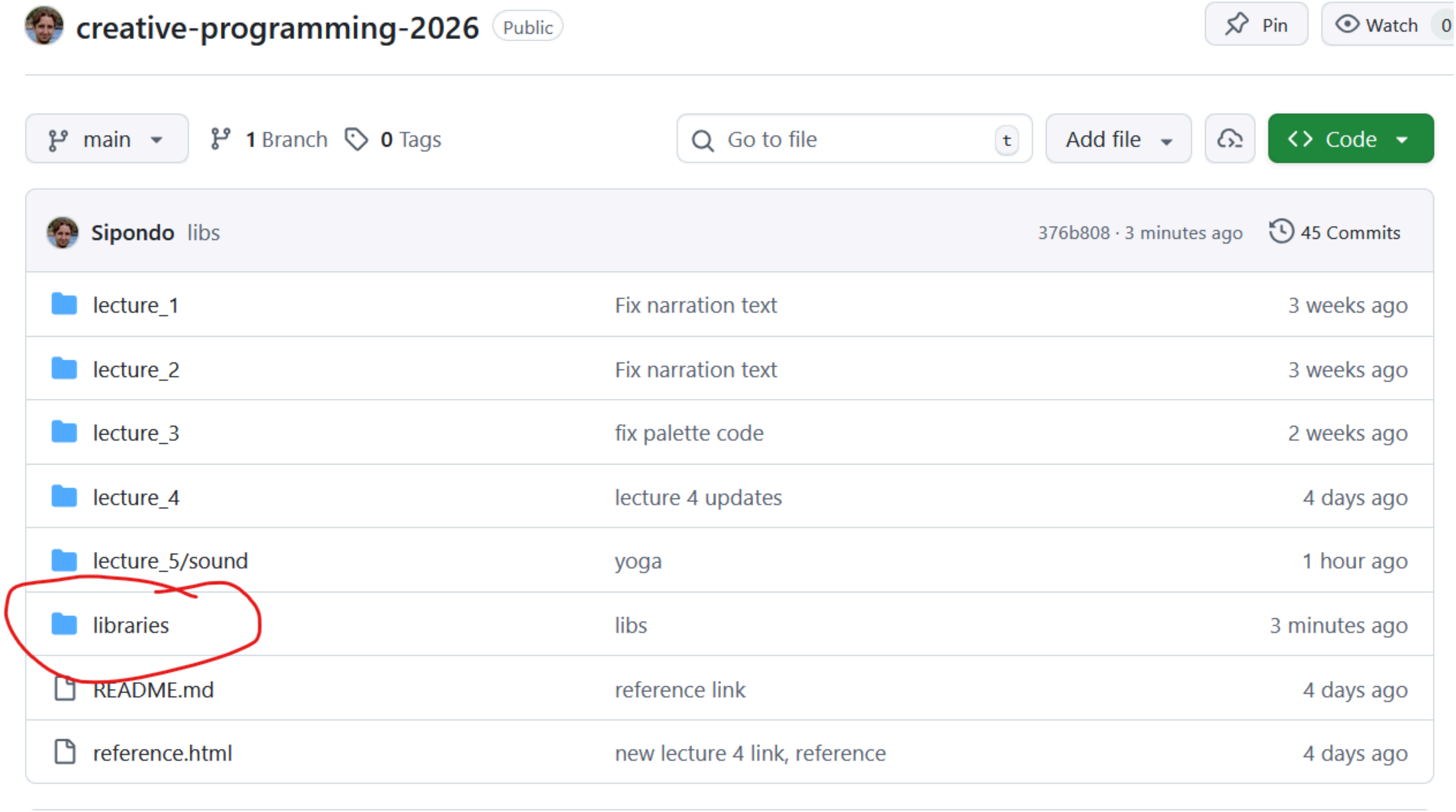
Downloading the p5js sound library (1)



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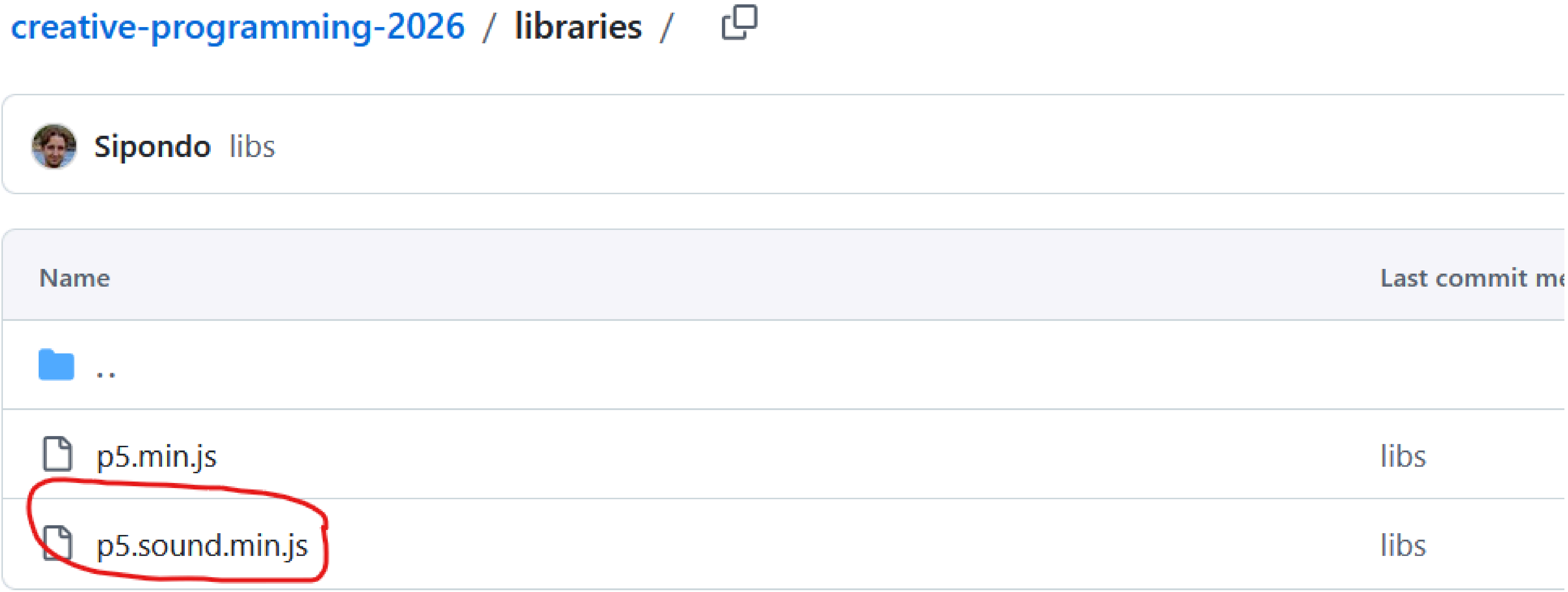
Go to the GitHub repository of the course and click on the `libraries` folder

Downloading the p5js sound library (2)



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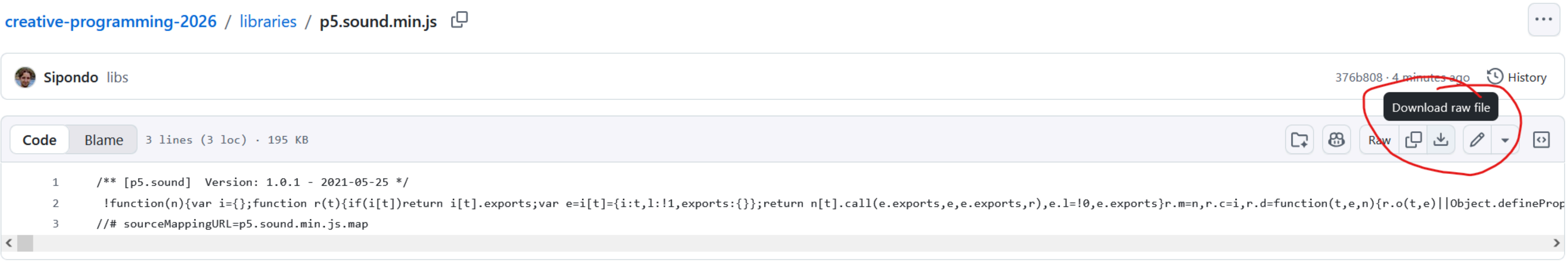
Open the `p5.sound.min.js` file

Downloading the p5js sound library (3)



Download the raw file

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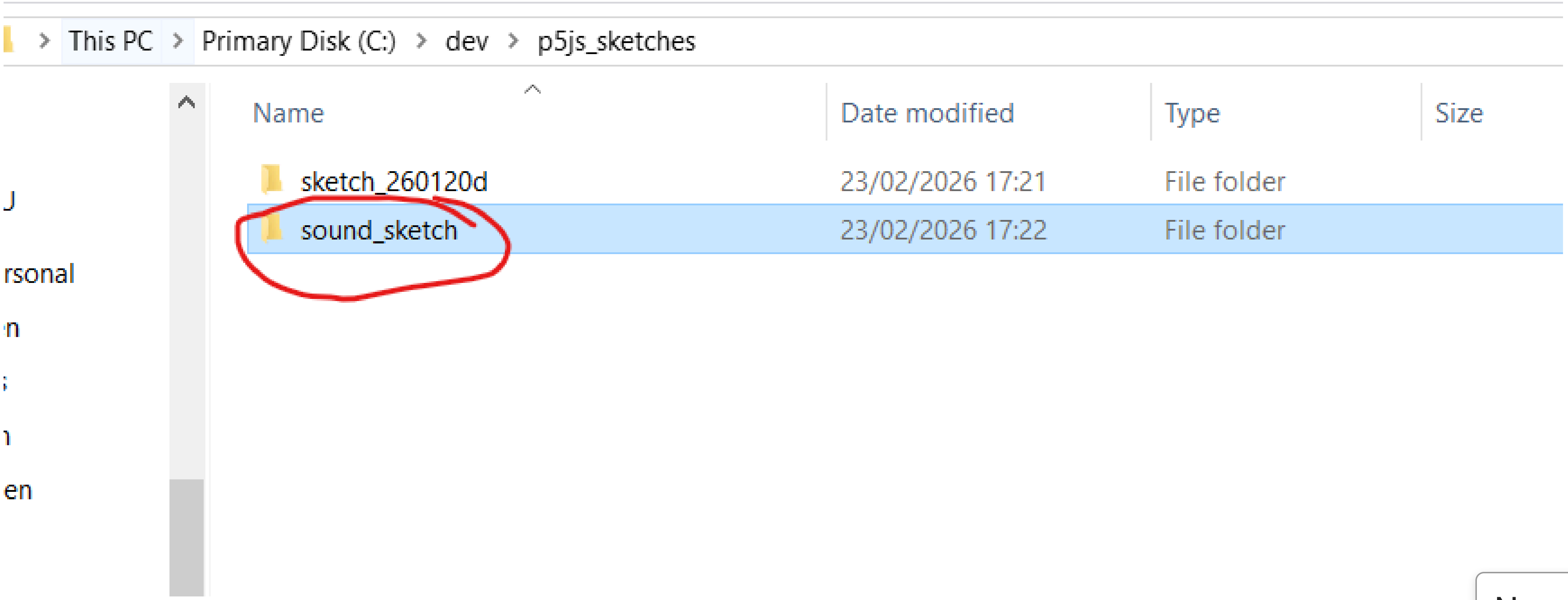
Putting the p5js sound library in your sketch (1)



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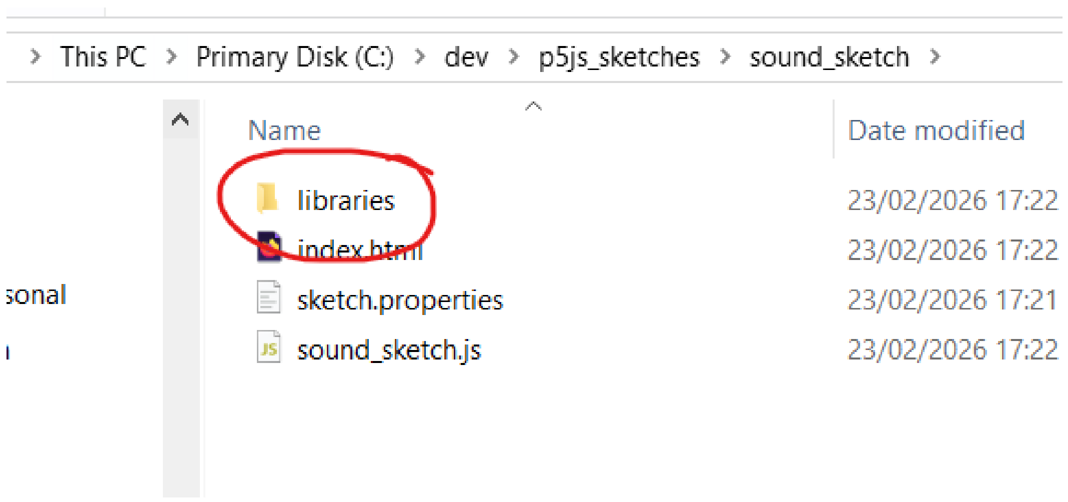
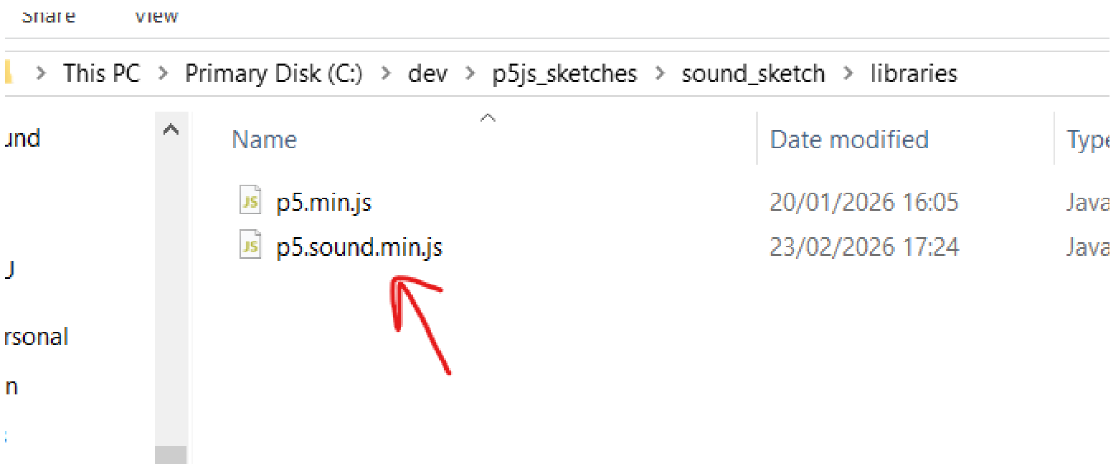
Make your (new) sound sketch and go to its folder

Putting the p5js sound library in your sketch (2)





Go to the libraries folder in your sound sketch and put the `p5.sound.min.js` file there

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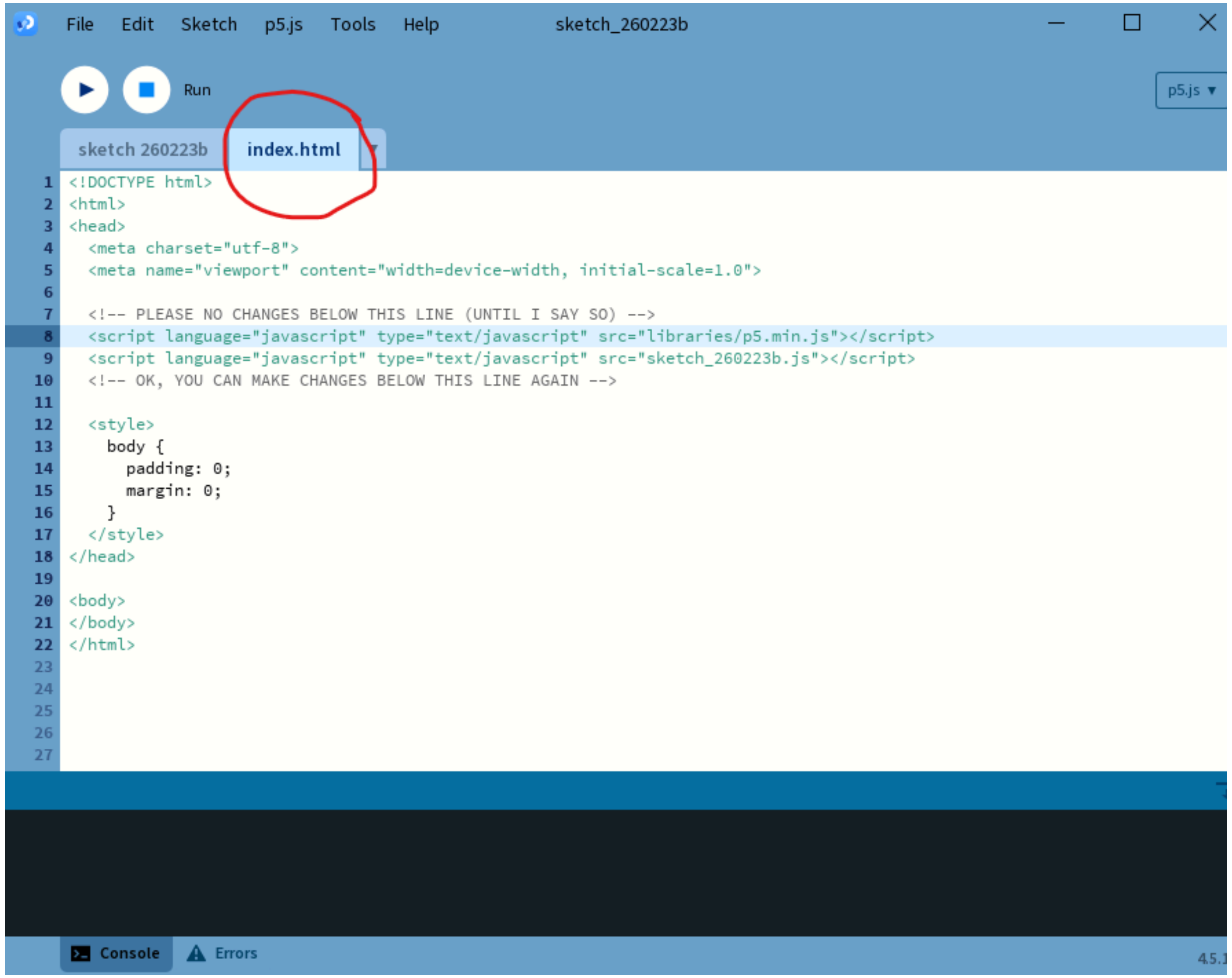
Setting up your sketch HTML (1)



In Processing, open up the `index.html` file of your sketch

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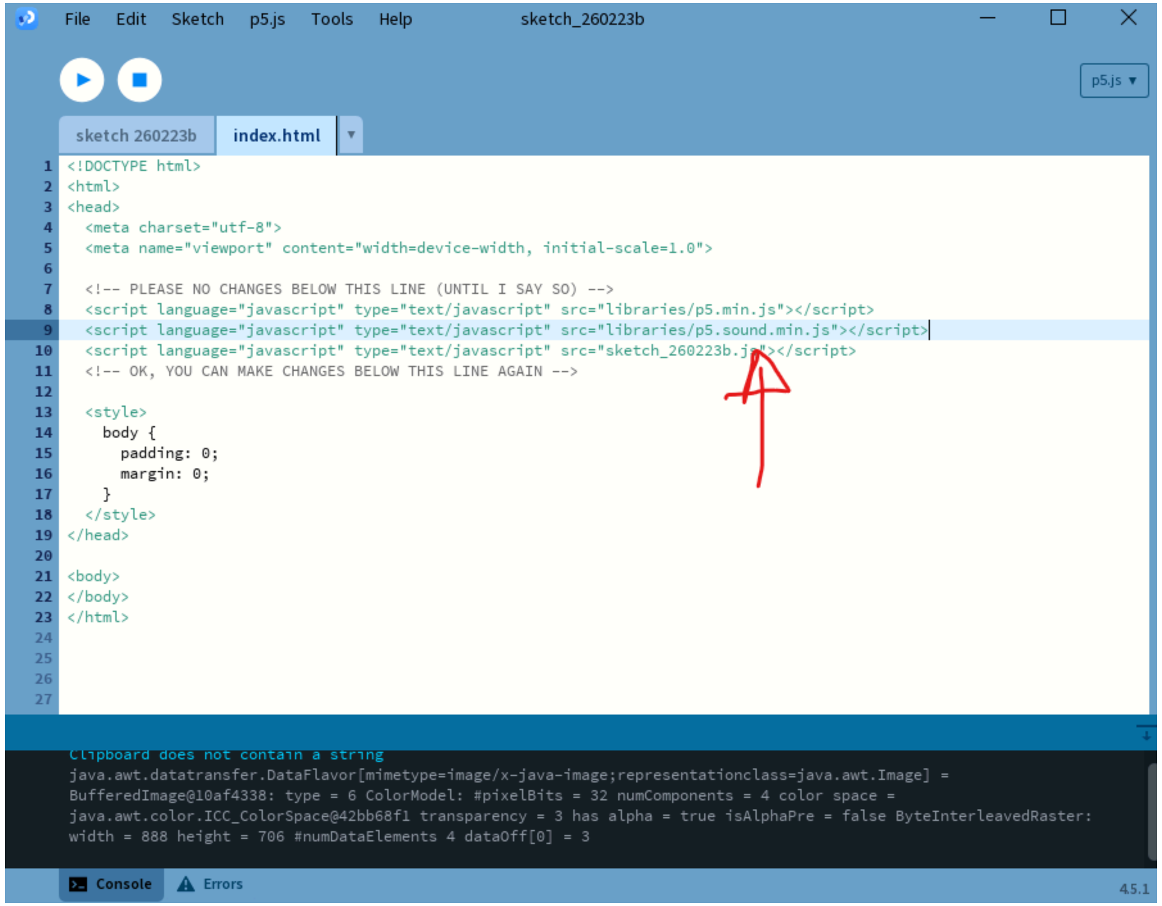
Setting up your sketch HTML (2)

Add the following to your index.html:

```html
<script language="javascript" type="text/javascript" src="libraries/p5.sound.min.js"></script>
```



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Setting up audio in your sketch

- Browser makers (Google, Mozilla, Apple, etc.) do not like it when websites play sound without the user's consent.
- To prevent this, they require that audio can only be played in response to a user interaction (e.g. a mouse click or key press).
- We have to call the `userStartAudio()` function in a user interaction event (e.g. `mousePressed()`) to enable audio in our sketch.

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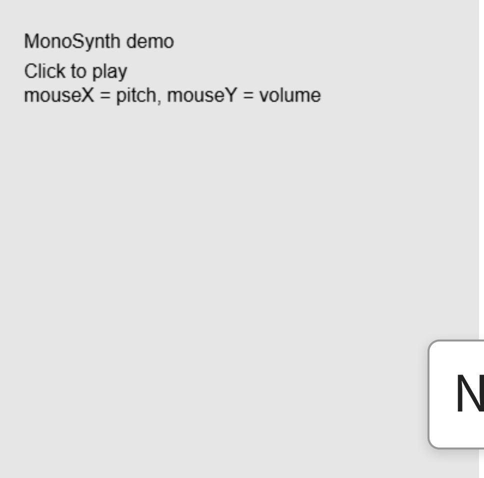
Playing a note

- We can play single notes by specifying their note name (e.g. “A4”) or their frequency in hertz (e.g. 440).
- `play(note, volume, delay, time)` takes four arguments to specify the pitch, volume (0-1), delay before the note starts (in seconds), and duration of the note (in seconds).

```
let mono;

function setup() {
  createCanvas(400, 400);
  mono = new p5.MonoSynth(); // Build the synthesizer. Don't forget "new"!
}

function draw() {
  background(230);
  textSize(16);
```



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Reacting to mouse position

- We can use frequencies to play notes that react to the mouse position.

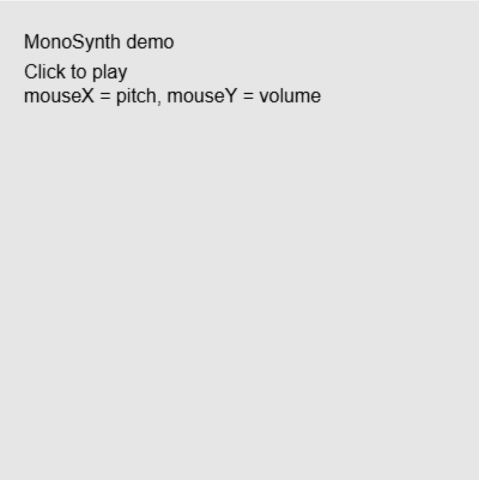
```
let mono;

function setup() {
  createCanvas(400, 400);
  mono = new p5.MonoSynth(); // Build the synthesizer. Don't forget "new"!
}

function draw() {
  background(230);
  textSize(16);
  text("MonoSynth demo", 20, 40);
  text("Click to play", 20, 65);
  text("mouseX = pitch, mouseY = volume", 20, 85);
}

function mousePressed() {
  userStartAudio(); // Start the audio context on a user gesture (required in some browsers)

```



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Playing multiple notes

- Multiple notes can be played with `p5.PolySynth()`
- Check out a website for this (e.g. chords) https://www.michael-thomas.com/music/class/chords_notesinchords.htm

```
let polySynth;

function setup() {
  createCanvas(400, 400);
  background(220);
  textSize(20);
  text('Click to play', 20, 20);

  polySynth = new p5.PolySynth(); // Synth that can play multiple at once
}

function mousePressed() {
  userStartAudio();
```

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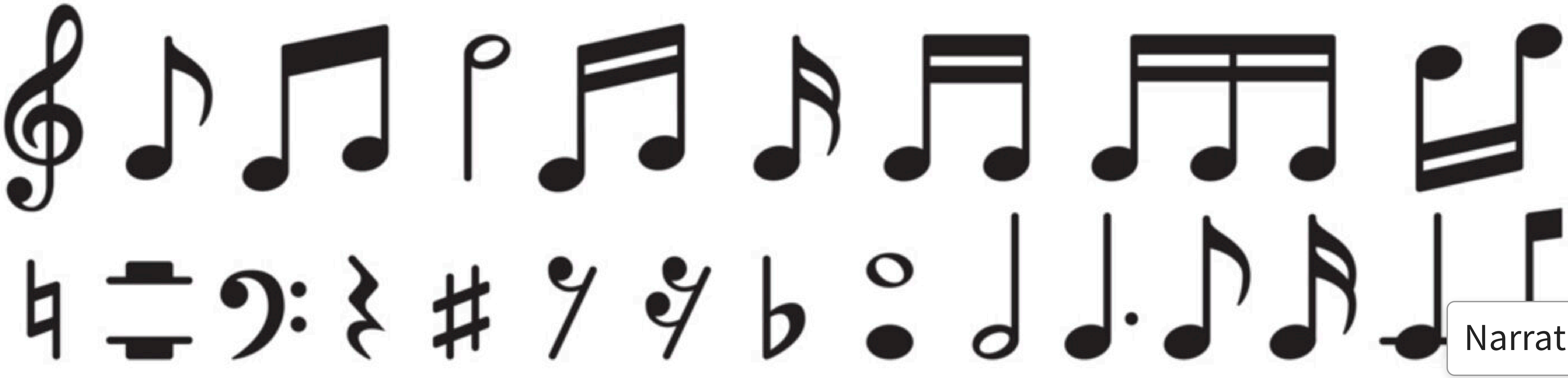
Synthesizing!

Class Assignment: Melody

1. Set up a synthesizer in p5js and play a note when a key is pressed.
2. Store a melody of notes in an array and play it back when a button is pressed instead.
3. By using a polySynth, can you play a melody of chords instead of notes?



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Playing sounds

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Sound files

- Of course we can also play sound files instead of synthesizing sounds ourselves.
- p5.js provides a `loadSound()` function to load sound files and a `play()` method to play them.
- You can stop, pause, and loop sounds as well using the `stop()`, `pause()`, and `loop()` methods.
- `setVolume()` sets the volume of the sound, similar to how we can set the volume of a synthesizer note.

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The return of preload

- Just like with images, we need to use the `preload()` function to load sound files before the sketch starts.
- This ensures that the sound files are available when we want to play them.
- Remember! Loading files can be done with:
  - A website link (such as in the slides)
  - A local file in the same directory as your sketch (e.g. `cymbal.wav` if you have the file in the same folder as your sketch)

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Sound example

- Let us look at an example of loading and playing sound files in p5.js.

```
let cymbal;
let drum;
let snare;

function preload() {
  cymbal = loadSound( // Load the three sounds. You should prefer using local files, but here we load from a URL for ease of sharing
    "https://rawcdn.githack.com/Sipondo/creative-programming-2026/793c774ec9b9341473cd847312e7b6e21afcd410/lecture_5/sound/cymbal.wav"
  );
  drum = loadSound(
    "https://rawcdn.githack.com/Sipondo/creative-programming-2026/793c774ec9b9341473cd847312e7b6e21afcd410/lecture_5/sound/drum.wav"
  );
  snare = loadSound(
    "https://rawcdn.githack.com/Sipondo/creative-programming-2026/793c774ec9b9341473cd847312e7b6e21afcd410/lecture_5/sound/snare.wav"
  );
```

Keys: C = cymbal, D = drum, S = snare

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Finding sounds

- There are many websites where you can find free sound files to use in your projects, such as:
  - Freesound
  - Free Music Archive
  - SoundBible

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Looping and pausing

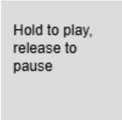
- If you want to have a sound repeat, use `loop()` instead of `play()`.
- You can pause and resume the sound using `pause()` and `play()`.
- Finally, you can stop the sound completely with `stop()` .

```
let soundFile;

function preload() {
  soundFile = loadSound("https://rawcdn.githack.com/Sipondo/creative-programming-2026/7bcbf236e2ca4d161acb370092f669ce38cd543a/lecture_5/sound/yogabells.wav");
}

function setup() {
  createCanvas(100, 100);
  background(220);
  text("Hold to play, release to pause", 10, 20, width - 20);
}

```



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Recording

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Microphone input

- Sound goes both ways! We can also capture sound using the microphone.
- In p5js, we can use the `p5.AudioIn()` class to access the microphone and capture sound input.
- We have to make a new `p5.AudioIn()` object and call the `start()` method to begin capturing sound.

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S(l)ide note

- Microphone input and interactive slides are not the best of friends.
- Some of the examples may not work properly in the slideshow, but you can copy the code into your own sketch to test it out.

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Volume levels

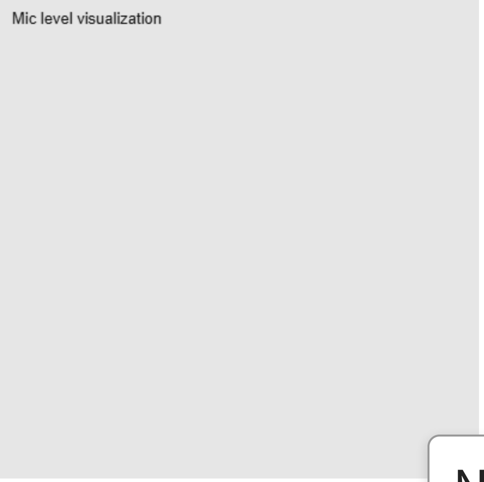
- The `getLevel()` method of the audio input object returns the current volume level of the sound being captured by the microphone.
- This can be used to create visualizations or interactive elements that respond to sound input.

```
let mic;

function setup() {
  createCanvas(400, 400);

  // Create and start the microphone input
  mic = new p5.AudioIn();
  mic.start();
}

function mousePressed() {
  userStartAudio(); // Start the audio context on a user gesture (required in some browsers)
```



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Mapping the microphone

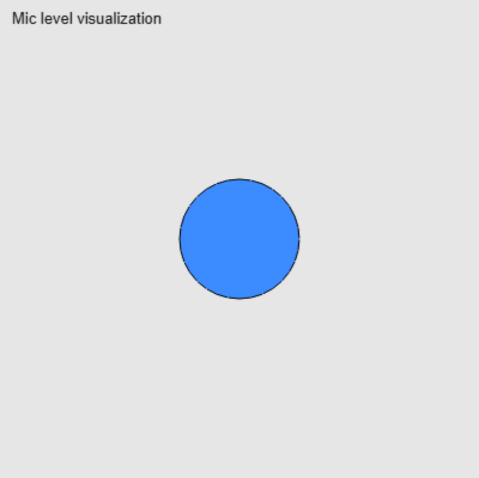
- The values returned by `getLevel()` are typically between 0 and 1.
- We can use the `map()` function to scale the volume level from the microphone to a range that is suitable for our sketch.

```
let mic;

function setup() {
  createCanvas(400, 400);

  // Create and start the microphone input
  mic = new p5.AudioIn();
  mic.start();
}

function mousePressed() {
  userStartAudio(); // Start the audio context on a user gesture (required in some browsers)
}
```



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Easing the microphone

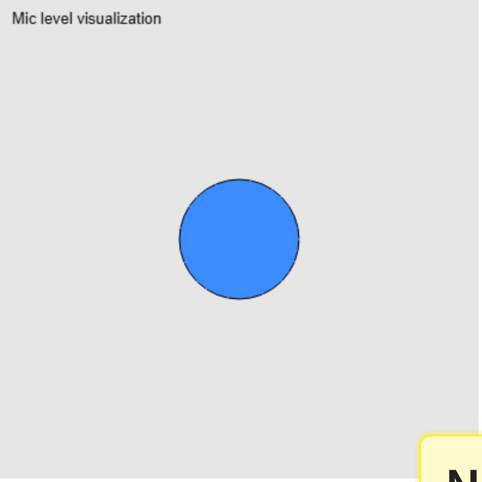
- The microphone values are very jittery and can change rapidly, which can make it difficult to create smooth visualizations or interactions.
- Easing can get rid of the jitter and create smoother transitions between values.

```
let mic;
let diameter = 50; // Initial diameter for the circle

function setup() {
  createCanvas(400, 400);

  // Create and start the microphone input
  mic = new p5.AudioIn();
  mic.start();
}

function mousePressed() {
```



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Sound as vibrations

- Microphone input does not just have to be sound
- Some media (such as games) use super high volume levels to check whether users blow into the microphone

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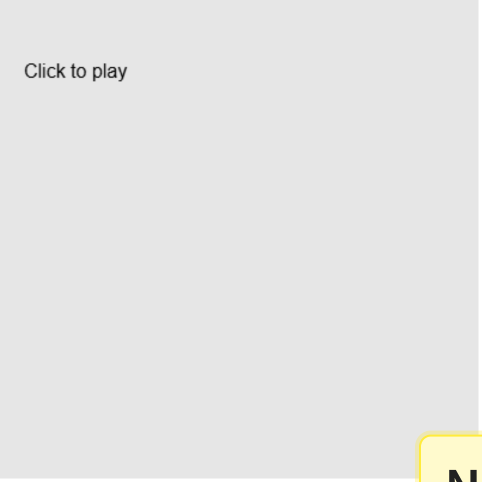
Microphone-controlled synthesizer

- We can combine the microphone input with a synthesizer to create a sound-reactive instrument.
- For example, we can use the volume level from the microphone to control the frequency or amplitude of a synthesizer note.

```
let mono;
let diameter = 50; // Initial diameter for the circle

function setup() {
  createCanvas(400, 400);
  mono = new p5.MonoSynth(); // Build the synthesizer. Don't forget "new"!
  mic = new p5.AudioIn(); // Create an audio input
  mic.start(); // Start the audio input
}

function mousePressed() {
  userStartAudio(); // Start the audio context on a user gesture (required in some browsers)
```



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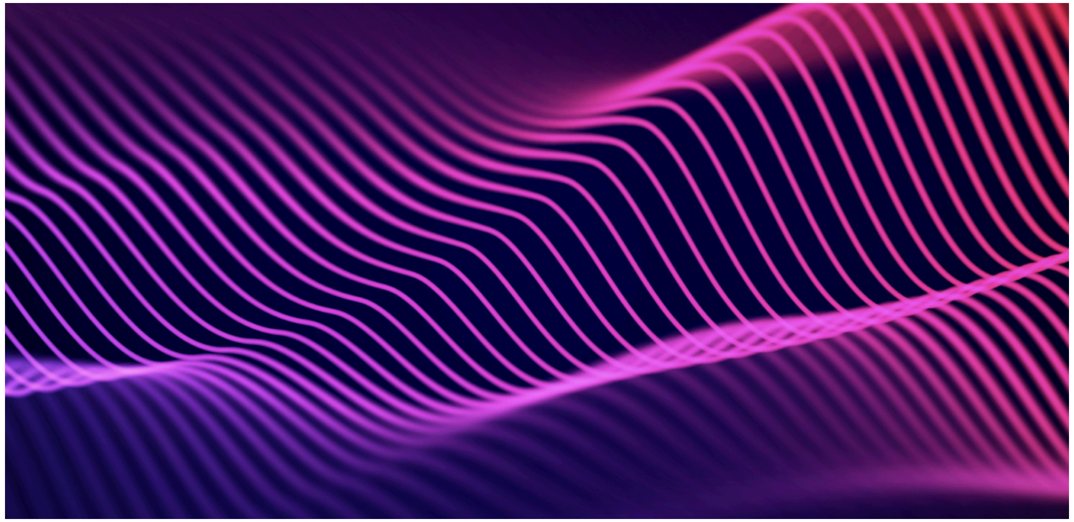
Sound reactive

Class Assignment: Sound Reactive

1. Set up a microphone input in p5.js and use the volume level to control a visual element (e.g. the size of a circle). You can also modify an old sketch!
2. Go on one of the free sound websites and find a sound file that you like. Load it into your sketch and play it when a button is pressed.
3. Can you make it so that the sound file loops as long as there is sound over the microphone?



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